

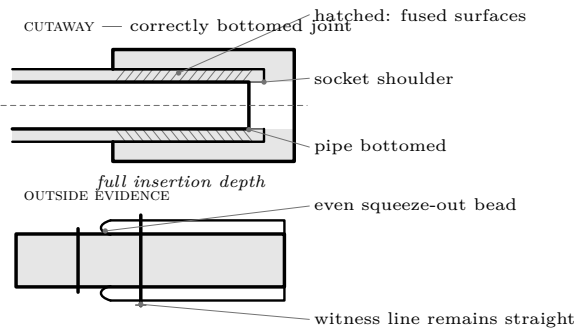
Craft 4: Solvent Weld

It is not glue. Nothing about it behaves like glue.

THE STANDARD

Every joint shows a **continuous even bead** of cement squeezed out all the way round the socket mouth, the pipe is bottomed against the shoulder, the witness mark you drew lines up, and the date is written on the part. A joint with a gap in that bead is a joint with a gap in the weld, and there is no way to fix it afterwards — you cut it out and start again.

What is actually happening

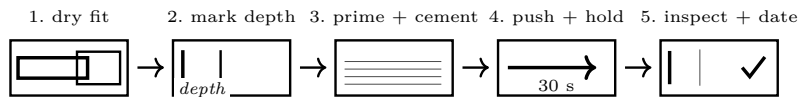


Primer	A solvent that softens and swells the surface of both parts and cleans them at the same time. It is doing chemical work, not cosmetic work. Skipping it is the single most common cause of a joint that fails under pressure.
Cement	More solvent, plus dissolved PVC to fill the gap. It dissolves the softened surfaces into each other.
The weld	As the solvent evaporates, the two dissolved surfaces re-solidify as one continuous piece of plastic. A correct joint is not two parts stuck together — it is one part. That is why a good weld is stronger than the pipe.
Which means	Clamping does nothing. More cement does not mean more strength. And you cannot test a joint by pulling on it, because the failure you care about happens at pressure, not in tension.
Time is a real ingredient	The solvent has to leave. Until it does, the joint is soft. This is why cure time is not a suggestion.

Method

1. **Dry-fit first, and mark it.** Push the pipe in without cement. It should go in most of the way by hand and need real effort for the last part. Draw a **witness mark** — a single line across the joint, from pipe onto fitting — and a rotation alignment mark if the fitting's orientation matters.
2. **Check the temperature.** Below about 5 °C the solvent barely works and cure times lengthen enormously. Above about 35 °C it flashes off before you can assemble. Work in the middle.
3. **Clean and dry both surfaces.** Wipe off swarf, dirt and any moisture.
4. **Primer on both,** socket first then pipe, covering the full insertion depth. It should look wet and slightly dulled.
5. **Cement on both,** while the primer is still wet. A full even coat on the pipe, a lighter one in the socket — too much in the socket pools at the shoulder and can hydraulic-lock the joint short of bottom.
6. **Insert with a quarter turn.** Push straight in, twisting a quarter turn as you go, until it bottoms. The twist distributes the cement; the push is what makes the weld. Stop turning once it is bottomed.
7. **Hold for 30 seconds.** A solvent joint actively pushes itself back out while the softened plastic relaxes. If you let go early it will creep out a few millimetres and you will never know how far.
8. **Wipe the excess,** check the bead all the way round, and **write the date on the part.**

One-way sequence — stop if evidence fails



Before opening primer, measure and record the dry insertion depth: _____ mm. Transfer that distance to the pipe as a depth line. After the 30-second hold, measure the remaining distance from socket mouth to that line: _____ mm. It must match the dry-fit relationship, and the alignment witness must remain continuous.

INSPECTION DECISION WHILE THE EVIDENCE IS VISIBLE

Ask: Is the pipe at the recorded depth? Is the witness line aligned? Is the squeeze-out bead continuous through a full turn? Is the cure date legible? **Accept** only four yes answers. A smear may be wiped, but a depth error, opened witness line, dry gap, contaminated surface, or missed assembly window cannot be repaired with more cement. Mark the joint rejected, cut it out when safe to handle, and remake it with new fitting and pipe ends.

You should see: A continuous ring of cement squeezed out at the socket mouth, all the way round with no gaps, and your witness mark aligned. If the mark has moved apart, the pipe did not stay bottomed — cut the joint out.

Cure time

Stage	Roughly	What it means
Set	30 s – 2 min	It will not pull apart. It is nowhere near strong.
Handling	15 – 30 min	You may move the assembly gently. Do not stress the joint.
Working	several hours	Suitable for low-pressure plumbing.
Full cure	48 hours	The figure this project uses. Larger diameters, thicker walls, cold weather and high humidity all make it longer, never shorter.

FORTY-EIGHT HOURS, AND WRITE THE DATE ON IT

The joint feels finished after twenty minutes. It is not. Solvent is still leaving the weld, and its pressure strength climbs for days. A chamber fired at four hours can fail at a pressure it would have shrugged off at forty-eight. Writing the date on the part in marker is how you stop a confident guess from substituting for a fact — and it takes three seconds.

Failure modes, and how each one looks

Mistake	What happens
No primer	The surfaces never soften, so the cement acts as an adhesive film instead of a solvent weld. Holds under hand pressure, fails under gas pressure. Invisible from outside.
Too little cement	Dry patches with no weld at all. Look for gaps in the squeeze-out bead — that is what the bead is for.
Too much cement	Pools at the shoulder, can stop the pipe bottoming, and in thin sections the solvent can soften more wall than intended.
Let go too early	The pipe creeps out. The witness mark catches this and nothing else does.
Not bottomed	Less weld area than designed and a step inside the bore. Dry-fitting first is what prevents it.
Cold or damp	Slow, weak cure. Wait for better conditions rather than hoping.
Fired before cure	The one on this list that hurts somebody.

Check it yourself. Run a fingertip round the socket mouth. You are feeling for a continuous ridge of squeezed-out cement with no gaps. Then check the witness mark is still aligned, then find the date you wrote and count forward forty-eight hours. Three checks, ten seconds, and they are the only evidence you will ever get about the inside of that joint.

You should see: Final proof before any use: cure date _____; earliest permitted inspection date/time _____. After the full cure, a second person verifies the depth line, witness alignment, complete bead, and elapsed time. Maker _____ checker _____. This is an inspection record, not permission to shorten the stated cure.

Practise on offcuts first

Weld three or four scrap joints before touching a real part, and then cut two of them in half lengthways with a saw. You will see immediately whether your technique produced a continuous fused zone or a patchy one — and you will never again wonder what is happening inside a joint you cannot see. Keep the best section on the bench as your reference for what good looks like.

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